

Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

WORK EXPERIENCE

PERFOMAX AI Advisor 2026–present

NATIONAL UNIVERSITY OF SINGAPORE Postdoc under Kostya Novoselov 2022–present

Leadership

- Program Chair for the [AI4X 2025/6 conferences](#).
- Main organizer of the [ICLR 2025 workshop on multiscale machine learning](#).

Wyckoff Transformer for generating symmetric crystals [ICML 2025](#) · [code](#)

- Developed a coordinate-free permutation-invariant autoregressive Transformer encoder model for material generation and property prediction based on symmetry inductive bias
- Increased materials discovery yield (S.U.N.) by 1.24x
- Accelerated inference throughput by 4 orders of magnitude, dropping generation costs to 0.05 GPU ms per structure.
- Implemented the model in PyTorch; production-ready package published on [PyPI](https://pypi.org/project/wyckoff-transformer/)
- Orchestrated large-scale (10k structures) DFT computations with VASP at an HPC cluster
- Led a team of 6

CERN [YANDEX → HSE UNIVERSITY] Researcher 2014–2022

[2025 Breakthrough Prize in Fundamental Physics](#) as a member of LHCb.

Generative models uncertainty estimation [paper](#) · [conference](#) · [code 1](#) · [code 2](#)

- Developed the first methods for estimating uncertainty of conditional GANs.
- Led a team of 3 students.

Generative models for fast simulation [paper 1](#) · [paper 2](#) · [paper 3](#) · [code](#)

- Developed a GAN-based machine learning model for high-fidelity simulation of a Cherenkov detector trained on tabular data.
- Achieved approximately 10^5 speed-up compared to the ab-initio simulator.

EDUCATION

Higher School of Economics (HSE University) 2016–2020
PhD in Computer Science, supervisor [Andrey Ustyuzhanin](#); [Machine Learning for particle identification in the LHCb detector](#).

Sapienza – Università di Roma 2016–2020
PhD in Physics (double degree with HSE), supervisor [Barbara Sciascia](#).

Product Management course at Yandex 2018
Focused coursework in product strategy and execution.

Yandex School of Data Analysis 2013–2015
Master's level CS course covering algorithms, machine learning, deep learning, and distributed systems.

Moscow Institute of Physics and Technology
MS in Physics (2014–2016): Optimisation of data processing of the LHCb experiment.
BS in Physics (2010–2014): Study of the quantum states of the electrons in nonideal plasma and selected molecules using wave packet molecular dynamics with packet splitting.

MENTORSHIP

- Mentored 9 students ([1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#)), 3 interns, and student workshop projects ([1](#), [2](#)).
- Student council member from 2013 to 2016, leading several dormitory-scale IT projects.
- Co-PI of a \$3.4m AI Singapore grant on multiscale machine learning.

TEACHING & OUTREACH

- Pitch research to the general public through major newspaper coverage ([1](#), [2](#)), blog posts ([1](#), [2](#)), a [science slam](#), interviews ([1](#), [2](#)), and a public [talk](#).
- Lecturer in the [Coursera Advanced Machine Learning specialisation](#), which is [award-winning](#).
- Delivered more than 20 [conference talks](#).

SERVICE

- Reviewer for RSC Advances, Machine Learning: Science and Technology, and AI for Accelerated Materials Design workshops at NeurIPS and ICLR (2023–2025).
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

OTHER

- Aspiring stand-up comedian; bombed more than 20 times at open mics.
- International Junior Science Olympiad 2008 in Korea, silver medal.

SKILLS

- **ML & AI** – Structured, non-text, domain-specific data
- **Python** – Research coding
- **PyTorch** – Deep learning
- **Research Writing & Speaking**
- **Leadership & Mentorship**

SELECTED PUBLICATIONS

1. **Kazeev, Nikita**, and Ian Babich. [Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments](#). Forecasting as a New Frontier of Intelligence Workshop at ICML 2026
2. **Kazeev, Nikita**, and Phan Bui Nhat Huyen. [Position: Align AI to Our Aspirations, Not Our Flaws](#). Pluralistic Alignment Workshop at ICML 2026
3. Kazeev, Nikita and Ustyuzhanin, Andrey. [The Geometry of Reasoning Failure: Predicting Agent Errors from Semantic Scale Trajectories](#). Statistical Frameworks for Uncertainty in Agentic Systems Workshop at ICML 2026
4. Nian Liu, **Nikita Kazeev**, Stephen Gregory Dale, Artem Maevskiy, Yuwei Zeng, Ryoji Kubo, Pengru Huang, Thomas Laurent, Yann LeCun, Kostya S. Novoselov, and Xavier Bresson. [Crys-JEPA: Accelerating Crystal Discovery via Embedding Screening and Generative Refinement](#). arXiv preprint arXiv:2605.14759, May 2026.
5. Alexandre Duval, Siddharth Betala, Samuel P. Gleason, Andy Xu, Georgia Channing, Daniel Levy, Ali Ramlaoui, Clémentine Fourrier, Chaitanya K. Joshi, **Nikita Kazeev**, Sékou-Oumar Kaba, Félix Therrien, Alex Hernández-García, Rocío Mercado, N M Anoop Krishnan. [LeMat-GenBench: Bridging the gap between crystal generation and materials discovery](#). AI for Accelerated Materials Design - NeurIPS 2025 / arXiv preprint arXiv:2512.04562, October 2025.
6. **Kazeev, Nikita**, et al. [WyckoffTransformer: Generation of Symmetric Crystals](#). ICML 2025, May 2025.
7. Derkach, D., **Kazeev, N.**, Ratnikov, F., Ustyuzhanin, A., & Volokhova, A. (2019). [Cherenkov detectors fast simulation using neural networks](#). *Nuclear Instruments and Methods in Physics Research Section A*. Alphabetic order, I'm the corresponding author.